Owner checkpoint | 13 October 2026

- The Linux workload, UART protocol and FreeRTOS supervisor are implemented; inference is an OS workload.
- Retained evidence covers Pi software timing, physical MCU transitions and analyzer captures, with explicit exclusions.
- Immediate work: document the rig, verify the review fixes, restore clean power, then rerun under v2 rules.

The code is ahead of the clean physical evidence.

Source: P_PROGRESS.md; results/README.md; docs/STAGE_REPORT.md

System map | execution and ownership

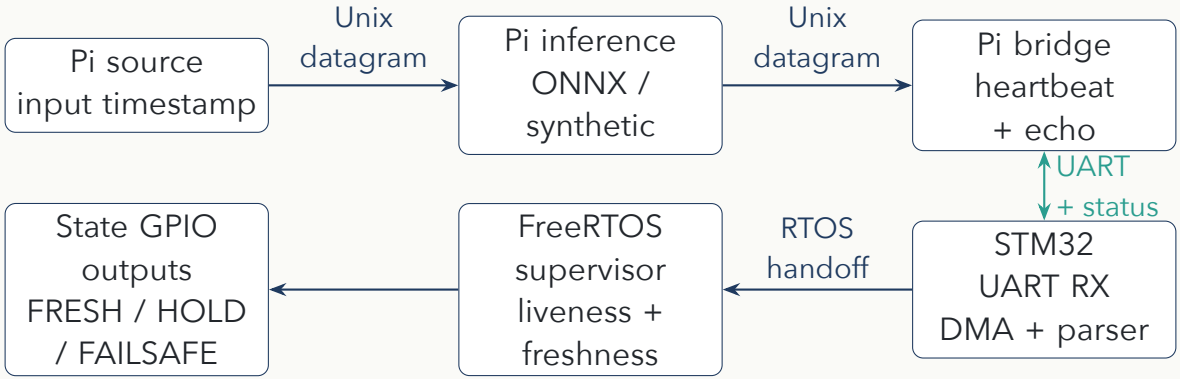


Figure 1: data flows to the supervisor; echo/status return to the Pi.

Figure 1 locates the decision on the MCU. Mac: builds, ST-LINK and analyzer; Pi: Linux processes and stressors. PTY runs replace the physical UART/MCU path.

An alive bridge and a fresh inference result are separate conditions.

Source: docs/STAGE_REPORT.md, What was built; linux/README.md; stm32/firmware/README.md

Timeline | foundations, 5–10 October

- 5 October: project scope, heterogeneous architecture and environment plan documented.
- 6 October: rig plan and packaged real-time Ubuntu test profile P3 decision; P3 is not a completed measurement.
- Early roadmap snapshot (a22fa2b): A2 clock math pending. Later 10 October: A2 implemented (e456e6b), integer-offset / zero-spread review fix (e3b81ab); A3/A4 also landed.
- Portable clock math is implemented; independent physical clock calibration remains open. Full Linux pipeline landed 11 October (fc37e3b).

Read the stage report for current state, the roadmap for sequencing.

Source: PROJECT.md; docs/ENVIRONMENT_AND_HARDWARE.md; roadmap snapshot; git e456e6b / e3b81ab / f19d05f / bc07462 / fc37e3b

Timeline | implementation and integration, 11–13 October

- 11 October: feat/pi-timing-tools and feat/inference-baseline; cyclictest correction, live power logs, UART return path and PTY campaign tools.
- 12 October: input-latest E3b, HSE clock, DMA RX, physical E4/F5, real UART E3, analyzer captures and stage report.
- 12 October: component branches merged into release/os-final; cleanup commit 5575cef unifies tests and indexes retained results.
- 13 October: 5e3f461 fixes all 11 review findings and records v1/v2 validity limits. PR #6 release/os-final is open; #3–#5 superseded (brief snapshot).

Merge status does not certify the new firmware on the rig.

Source: git log: 172e415, fdeaa6f, 241d36f, 5575cef, 5e3f461; P_PROGRESS.md

Status board | software and protocol

- IMPLEMENTED + TESTED (host): shared parser/CRC, portable supervisor, restart/session checks, DMA cursor and tick diagnostics.
- IMPLEMENTED + TESTED (host): Linux source, inference, bridge, receiver and analysis/tooling; retained tests are historical checks.
- MEASURED: Pi inference and software/PTY campaigns under the stated v1 scopes; these are not physical UART AoI.
- FLAGGED: loaded input-latest and physical E3 comparisons. NOT DONE: clean physical E1/E2 and complete held-value AoI.

Host logic, target operation and accepted measurements are different milestones.

Source: docs/STAGE_REPORT.md; results/README.md; Makefile

Status board | firmware and rig

- VERIFIED ON RIG (narrow scope): earlier firmware exchanged status, echo and transition records; later analyzer captures are retained.
- MEASURED: accepted MCU-local E4/F5 transitions. FLAGGED: bring-up smoke, mixed faults and analyzer inventory.
- IMPLEMENTED + TESTED (host/cross-build): 11 review fixes. NOT DONE: flash and rig verification of this revision.
- NOT DONE: independent clock calibration, stack margin qualification, calibrated transit/rearm margins and hardware watchdog integration.

A previous rig run cannot verify a newer binary.

Source: stm32/firmware/README.md; results/README.md; review commit 5e3f461

Evidence rules | v1 history and v2 reruns

- MEASURED under v1: E1/E2/E3 PTY 45/45 accepted. v2 replay: 0/45, because RAW monitor timestamps were not retained.
- MEASURED under v1: cyclictest 15/15 per profile. v2 replay: 3/15 per profile; loaded runs lack stressor-liveness evidence.
- E4 stays 17/25 on replay; analyzer captures unchanged. Missing v2 evidence does not prove historical runs malfunctioned.
- v2 requires whole-window RAW monitor coverage, gaps ≤1.5 s, stressors alive and clock fits no older than 5 s.

Retain v1 observations with their scope; every new campaign uses v2.

Source: results/README.md, Validity rule versions; docs/STAGE_REPORT.md

Experiments | timer and inference baselines

- MEASURED: cyclictest timer wake-up latency, 30 runs, five loads × two profiles × three repeats; 120 s per run.
- MEASURED: explicit P0 OTHER and P1 FIFO policies checked on the measured thread. v2 retains only the six idle runs.
- FLAGGED mixed inference baseline: 21/24 accepted; three four-thread runs excluded for live under-voltage.
- FLAGGED initial inference collection: 22/24 excluded by original sticky-bit rules; no continuous live-power log. Superseded, not reclassified.

Timer wake-up and synchronous inference-call time measure different mechanisms.

Source: results/2026-10-11_cyclictest/SUMMARY.md; results/2026-10-11_inference_baseline/SUMMARY.md; results/README.md

Experiments | E1 contention and E2 scheduling

- E1 measures service/delivery under idle, CPU, stream, cache and memory-rate load; 15/15 accepted under v1.
- E2 compares scheduling profiles under matched loads; 18/18 accepted under v1. Applied policy and placement belong beside results.
- Service includes input waiting and preprocessing; delivery ends at a Pi PTY receiver. Neither is a physical MCU latency.
- INFERRED: memory contention and upstream queueing explain large tails; priority alone did not remove them. v2 cannot confirm these 33 historical runs.

Before rerunning, verify both the applied profile and the observer evidence.

Source: docs/STAGE_REPORT.md, E1 contention and E2 tuning; results/README.md

Experiments | E3 replacement and E3b input policy

- E3 changes bridge FIFO/latest policy; 12/12 accepted under v1, PTY only. Replacing after inference cannot clear its input backlog.
- INFERRED from advancing completions and zero bridge replacements: the overload queue was upstream of the bridge.
- E3b moves latest-value selection before inference; 12 runs, only 3 accepted. All six loaded plus three idle runs fail live power checks.
- FLAGGED loaded observations support debugging, not a clean benefit claim. Latest selection does not cancel inference already running.

Rerun input policy under clean power before claiming a physical freshness gain.

Source: docs/STAGE_REPORT.md, Freshness; results/2026-10-12_e3b_input_latest/

Experiments | physical E3 and bring-up

- FLAGGED physical E3: 18 runs, 0 accepted; 16 live-power flags, 9 loaded clock-fit failures and 14 missing transition records overlap.
- Clock residual RMS over 200 µs under memrate prevents clean mapped-time claims. Empty-event rejection has a documented validity limitation.
- FLAGGED bring-up: seven retained smoke directories, all excluded by summaries. DMA smoke6 has zero UART/CRC counters in its nominal 60 s run.
- FLAGGED separate real-UART rerun: 18 attempts, 0/18 valid; 18 power, 3 clock, 1 setting and 6 pipeline failures overlap. Six never opened a measurement window: no FRESH before startup deadline.

Missing FRESH root cause is UNKNOWN; require startup/rearm pilots (pp. 25, 29), beyond PSU replacement.

Source: results/README.md; docs/STAGE_REPORT.md; results/2026-10-12_e3_real_uart_rerun/

Experiments | E4 faults and F5 stale data

- Original E4: 25 planned entries; 20 performed, 17 accepted. F1 bridge kill, F2 bridge stop, F3 inference stop each 5/5; F4 FIFO hog 2/5; original F5 unperformed.
- Extra F4: 9 attempts, 5 accepted for short-window transitions; 3 startup failures and one RMS exclusion. No original recovery qualification.
- Separate F5: 5/5 accepted short-window stale-result transitions. SIGUSR1 suppresses inference while heartbeat continues.
- MEASURED MCU-local transitions are not independently calibrated safe-output bounds; observed maxima remain observations.

Keep acceptance denominator, exposure and transition reference together.

Source: results/README.md; docs/STAGE_REPORT.md, Physical faults / Extra F4

Analyzer evidence | later than the stage report

- FLAGGED inventory: 16 fault attempts, 15 captures, plus channel check; F1-r05 incomplete.
- F1, F2 and F4 captures compare GPIO marker, UART activity and state GPIO edges on one analyzer clock.
- Generated comparisons retain UNVERIFIED ON HARDWARE labels and differing MCU/analyzer reference events; no collection-wide promotion.
- The stage report predates these captures. Its pending-analyzer statements describe the earlier measurement scope.

Compare like reference events and retain each capture's qualification.

Source: results/README.md chronology note; tools/la/README.md; results/2026-10-12_logic_analyzer/

Rig | access, deployment and separation

- Last recorded OS Pi access: richard@192.168.50.2 over direct Ethernet; reconfirm reachability and identity. No Pi internet is assumed.
- Transfer Git bundles and ARM64 wheels from Mac. Initial bootstrap needs reachable apt sources; bundles/wheels do not provision system packages.
- Nucleo ST-LINK and FX2 analyzer belong to the Mac in the current runner; Pi owns UART and the fault-marker GPIO.
- UART serial console must be disabled before opening /dev/ttyAMA0. Direct Ethernet is management access; Pi–MCU data stays UART.

Reconfirm the actual rig before running old deployment notes.

Source: docs/CONDUCTOR_STATE.md; P_PROGRESS.md; tools/la/la_e4.sh; scripts/pi/UART.md

Rig | wiring and readiness gate

- Pi pin 8/GPIO14 TX → PA10/USART1 RX; pin 10/GPIO15 RX ← PA9/USART1 TX; shared ground.
- GPIO17/pin 11 → PA0/TIM2_CH1 is the sync route; GPIO27 marks faults. Confirm actual routing and analyzer channels in a rig photo.
- USART1 runs at configured 921600 baud; USART2/ST-LINK VCP is a separate debug path. HSE bypass depends on ST-LINK MCO.
- PLANNED gate: 5 V/5 A supply, cooling, console ownership, zero live flags, correct binary/hash and healthy FRESH startup.

Photograph and record the setup; configured values are not calibrations.

Source: stm32/firmware/README.md wiring; tools/la/README.md; P_PROGRESS.md

Tools | Pi provisioning and individual runs

Initial provision (Pi; apt network required)

```
sudo bash scripts/pi/bootstrap.sh
```

Requires reachable apt sources; installs packages/configures UART. Offline rig: audit dependencies first (README).

Metadata snapshot (Pi)

```
bash scripts/pi/collect_metadata.sh > meta.json
```

Writes platform, power and revision metadata to meta.json.

Detached command wrapper (Pi)

```
bash scripts/pi/run_experiment.sh check -- /usr/bin/true
```

experiments/runs/<UTC>_check/: metadata, logs, exit_code, validity, DONE.

Source: scripts/pi/bootstrap,collect_metadata,run_experiment.sh

Tools | pipeline and controlled faults

Single PTY pipeline (Pi)

```
bash scripts/pi/run_pipeline.sh --duration 60 --run-dir /tmp/pipe
```

/tmp/pipe/: config, CSV timestamps, counters and logs; stub by default.

Stop bridge briefly (Pi; substitute its recorded PID)

```
bash scripts/pi/fault_inject.sh stop "$BRIDGE_PID" 1000 2 /tmp/fault.csv
```

/tmp/fault.csv onset; stop/continue target process only.

GPIO sync helper (Pi)

```
python3 scripts/pi/uart_tools.py gpio 17 /tmp/sync.csv
```

/tmp/sync.csv RAW markers; Ctrl-C ends. fault mode is used by wrapper.

Source: scripts/pi/run_pipeline.sh; fault_inject.sh; uart_tools.py

Tools | software and timing campaigns

E1/E2/E3/E3B (Pi)

```
bash tools/experiments/e_campaign.sh ALL /tmp/e 3 60 --stage onnx --model
"$MODEL"
```

/tmp/e/: per-run pipeline/monitor/settings, validity; summary/. ALL follows runner cells.

Timing profile (Pi)

```
bash tools/timing/cyclictest_campaign.sh P0 /tmp/cyc 3 120
```

/tmp/cyc/: histogram, thread policy, power/liveness and summary; repeat with P1/new directory.

Python entry point behind shell wrapper

```
python3 tools/experiments/e_campaign.py E3B /tmp/e3b 3 60 --stage onnx
--model "$MODEL"
```

/tmp/e3b/: input-policy campaign. MODEL is an existing offline ONNX model.

Source: tools/experiments/e_campaign.sh,py; tools/timing/cyclictest_campaign.sh

Tools | physical campaigns and startup

Real-UART E3 (Pi; Mac reset helper required)

```
bash tools/experiments/e3_real.sh /tmp/e3real 3 60 --model "$MODEL"
--rearm-command "bash /home/richard/osbuild/rearm_via_mac.sh"
```

Setup Mac SSH/reset via REARM_SETUP.md first. /tmp/e3real/: pipeline, validity, summary; e3_real.py accepts same arguments.

E4 single stale-data trial (Pi)

```
bash tools/experiments/e4_campaign.sh /tmp/f5 1 --single F5 --rearm-command
"bash /home/richard/osbuild/rearm_via_mac.sh"
```

/tmp/f5/F5-r01/: power, fault/run JSON and pipeline; e4_campaign.py accepts identical arguments.

Startup helper (Pi; usually called by pipeline)

```
python3 tools/experiments/e3_startup.py /tmp/e3real/run/pipeline 30
```

Waits for post-reset FRESH event; writes fresh_start.json, errors on timeout/loss.

Source: tools/experiments/e3_real,e4_campaign.sh,py; e3_startup.py

Tools | Mac flashing and analyzer

Flash/reset ST-LINK (Mac with tools installed)

```
bash scripts/stm32/flash.sh openocd
```

Verifies stm32/firmware/build/firmware.elf and resets. reset.sh: bash scripts/stm32/reset.sh; no output file.

Capture fault with FX2 (Mac)

```
bash tools/la/la_e4.sh /tmp/la 5 F2
```

/tmp/la/: capture.sr/VCD, Pi logs, la_e4.csv/tex and optional timing PNG. Needs Mac ST-LINK / st-flash, FX2 / sigrok, Mac-to-Pi SSH and confirmed LA_PI / LA_PI_REPO. Reset is local.

Analyze retained/new capture copies (Mac)

```
python3 tools/la/la_analyze.py /tmp/la
```

/tmp/la/: comparisons and optional PNG. Analyze a copied capture tree to preserve originals.

Source: scripts/stm32/flash,reset.sh; tools/la/la_e4.sh,la_analyze.py

Tools | analysis entry points

One pipeline JSON report

```
python3 analysis/pipeline_summary.py /tmp/pipe > /tmp/pipe-summary.json
```

Reception age/latency distributions; not time-average held-value AoI.

E campaign and cyclictest summaries

```
python3 analysis/e_summary.py /tmp/e --outdir /tmp/e-summary
```

CSV/TeX + README in /tmp/e-summary. Timing: python3 analysis/cyclictest_summary.py /tmp/cyc --outdir /tmp/cyc-summary

Physical E4 and real-E3 summaries

```
python3 analysis/e4_summary.py /tmp/f5 --outdir /tmp/f5-summary
```

Fault CSV/TeX in /tmp/f5-summary. Real E3: python3 analysis/e3_real_summary.py /tmp/e3real → campaign/summary/.

Source: analysis/pipeline_summary,e_summary,cyclictest_summary,e4_summary,e3_real_summary.py

Pitfalls | policies, processes and worktrees

- cyclictest -p 0 can select FIFO rather than OTHER. Use explicit --policy=other/--policy=fifo and inspect measured thread sched_check.txt.
- pkill -f can match its own launching shell. Target recorded PIDs; validate ownership and use bounded stop/continue helpers.
- Worktree firmware build: set FREERTOS_REF to the existing reference kernel path. Do not initialize/reorganize reference code to repair a path.
- Cleanly rebuild configuration variants; stale objects after rapid configuration switches previously threatened reproducibility.

A command's requested setting is not proof of the applied setting.

Source: tools/timing/README.md; stm32/firmware/README.md; P_PROGRESS.md

Pitfalls | power, reason codes and clocks

- Recorded rig: 5 V/3 A; live under-voltage contaminates loaded comparisons. PLANNED: 5 V/5 A supply and continuous live/new-sticky checks.
- STATE_EVENT reason 9: source-age HOLD, reference last inference. Reason 3: heartbeat FAILSAFE, reference last heartbeat. Reason 10: source-age FAILSAFE.
- MEASURED accepted fault observations: about 175 ms after inference, 100 ms after heartbeat; earlier 98 ms HOLD heartbeat age was mislabelled. Different intervals, not bounds/thresholds.
- Keep Pi RAW and MCU timer domains separate; mapping needs fresh fit/RMS qualification. Internal tick self-check is not absolute calibration.

Read transition reason before choosing the reference timestamp.

Source: docs/STAGE_REPORT.md, Problems found; results/README.md; analysis/e4_summary.py

Next gate 1 | rig record and review-fix verification

- PLANNED: add a dated rig photo with board, supply, cooler, TX/RX/GND, ST-LINK and analyzer channels; confirm console ownership.
- Build exact revision 5e3f461 (or documented successor), record binary hash, configuration and FREERTOS_REF, then flash through Mac ST-LINK.
- Check five pipeline/firmware fixes: baud transit budget, partial-write expiry, signal-safe waits, bounded latest drain and input timestamp validation.
- Check six observer/tool fixes: fit age, monitor coverage/gaps, stressor liveness, decoder resync, CPU sample instants and failed-command invalidation.

Exit gate: all 11 fixes have recorded checks; new firmware reaches FRESH on rig.

Source: git show 5e3f461; stm32/firmware/README.md; P_PROGRESS.md

Next gate 2 | clean v2 evidence

- PLANNED: rerun timing, software policies, loaded E3b and physical E3 with 5 V/5 A PSU, matched workload and recorded cooling.
- Capture monitor RAW timestamps across the whole window, ≤1.5 s gaps, applied settings and stressor liveness. Preserve every failure.
- Physical pilot gate: verify reset and post-reset FRESH in idle + loaded cells, including both failed FIFO-input cells; retain startup failures. Fit age ≤5 s; RMS ≤200 µs.
- Target five independent valid repeats per cell. Extend F5 to original exposure/recovery; compare analyzer reference events and calibrate clocks.

Exit gate: valid-repeat target plus raw evidence, exclusions and scoped summaries.

Source: docs/STAGE_REPORT.md, Next steps; results/README.md; campaign summary/README.md

Next gate 3 | PR and final submission package

- PLANNED: reconcile the review-fix revision with PR #6 release/os-final; check CI, review findings and rig-verification record before merge.
- PR #3–#5 are superseded in the brief snapshot. Avoid integrating stale component branches over the release candidate.
- Package defense, report and owner deck with sources, PDFs, reproducible build commands, evidence inventory and review logs.
- Exit gate: fresh checkout builds; links and paths resolve; retained flags/v1-v2 limits survive packaging; final page review has no overflow.

Merge and submission are explicit completion gates, not inferred from a build.

Source: P_PROGRESS.md; submission/briefs/COMMON.md; git log release integration

WBR connection | roadmap A–F

- PLANNED roadmap: A Pi software → B external MCU → C physical experiments → D WBR shadow logging → E controlled integration → F article.
- Current evidence advances A–C, with power/clock/verification gates still open. Older roadmap completion markers are historical.
- Transfer timestamp discipline, liveness/freshness separation, upstream replacement and independent fault observation.
- Before D/E, review actual interface and fallback: reject stale Pi commands while preserving local balancing control. Re-measure all thresholds.

The method transfers; this rig's timing numbers and safe-output semantics do not.

Source: docs/ROADMAP_TO_WBROBOT_AND_ARTICLE.md; docs/STAGE_REPORT.md, Limits and transfer

Study pointers | mechanisms behind the checks

- Processes / scheduling / contention: long deck pp. 9, 10, 11, 82; learning-plan modules 0, 1, 4, 7.
- Queues / freshness: long deck pp. 12, 13; modules 5, 6 and 12.
- RTOS / supervision / clocks: long deck pp. 16, 14, 15; modules 8, 10, 11.
- PENDING owner lookup: exact Obsidian M-lessons UNKNOWN. Gate: map every mechanism above to a supplied/authorized note title + path; see README. No external vault searched.

Read the mechanism, then trace its implementation and one retained run.

Source: slides_course/main.tex frame sequence; docs/LEARNING_PLAN.md; docs/obsidian/2026-10-08-pi5-phase.md

First session | concrete checklist

- Confirm dedicated OS Pi identity and direct-link access; photograph the rig and install the clean supply/cooling.
- Build, hash, flash and validate the review-fix firmware; retain startup/status/counter logs and all 11 verification outcomes.
- Run idle + loaded v2 pilots, including input FIFO / bridge FIFO and latest. Gate: verified reset, post-reset FRESH and opened window; then audit power, coverage, settings, clocks.
- Archive raw runs and scoped summaries; resolve PR #6 gate, then rebuild all final deliverables and record their visual review.

Start with an accepted pilot; scale only after the evidence pipeline passes.

Source: P_PROGRESS.md; results/README.md; docs/STAGE_REPORT.md, Next steps